Venkata Lakshmi Parimala Pasupuleti

Software Engineer · Python · ML Pipelines · Feature Engineering · Model Deployment · Backend Systems · PostgreSQL · AWS · GCP · Real-Time Pipelines · A/B Testing · Full-Stack

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PROFESSIONAL SUMMARY

Software Engineer with 8+ years production backend and full-stack experience, and 2+ years building ML-powered systems end-to-end. Strong in Python, SQL, data pipelines, feature engineering, model deployment, and real-time event-driven architecture. Built AgentProbe: open-source ML evaluation infrastructure with model routing, 5-metric eval engine, A/B-style regression testing, and CI-integrated harness. Built CrashClarity: end-to-end ML pipeline over 104K records (Random Forest + Logistic Regression, 85%+ accuracy, feature engineering across weather/road/time variables). Deployed inference API (model serving) at GMU via PyTorch + HuggingFace. M.S. Data Analytics Engineering, George Mason University (GPA 3.87). — github.com/venkatalakshmiparimala/AgentProbe

WORK EXPERIENCE

Blue Horse Digital Sep 2026 – Present

AI/ML Engineer (Volunteer) · Remote

- Building ML models and data pipelines; surfacing insights from raw business data into application-facing APIs and reporting layers.

George Mason University Aug 2024 – Aug 2025

Research Associate

- **Built multi-source prediction ML pipelines** (Python, R) on WHO and UK Road Safety datasets across 190+ countries — $R^2 = 0.977$ regression accuracy; full pipeline from data collection and feature engineering through hypothesis testing and validation.
- **Fine-tuned a custom classification model** (HuggingFace Transformers + PyTorch); deployed inference API on Quart (async Python web framework) — model serving in production. Applied Isolation Forest for automated anomaly detection across 3 research data pipelines.
- Designed Tableau, ggplot2, and Quarto dashboards for faculty leadership; A/B-style evaluation frameworks across model iterations.

Entrata (PropTech / Real Estate SaaS) Sep 2018 – Aug 2024

Senior Software Engineer · Remote

- **Owned high-volume data pipelines in production for 7 years** — annual batch pipeline transforming federal source data into ~180,000 domain records powering eligibility, rent, and financial calculations for 100K+ housing units. Every release included baseline validation, reconciliation, and rollback preparation. Zero compliance violations.
- **Built real-time event-driven system (RabbitMQ)** for processing account state updates across thousands of records — retry logic, idempotency guarantees, fault-tolerant design, owned through production incidents. Optimized 200+ SQL queries (30% performance gain) through execution plan analysis, indexing, and partitioning.
- **Owned multi-tenant PostgreSQL schemas** for financial, compliance, and document workflows — schema design supporting data isolation across 100K+ units; performance engineering for high-throughput batch and real-time workloads.
- Shipped 20+ full-stack production modules (Python, JavaScript, PostgreSQL) across 120+ Agile sprints; microservices redesign (Docker) — 50% scalability improvement, 40% deployment time reduction.

Entrata India, Pune Jul 2017 – Sep 2018

Associate Software Engineer

- Built backend modules (Python, Java, PostgreSQL) across data pipelines and compliance workflows; led AWS S3 cloud migration; redesigned document storage architecture for hundreds of thousands of records.

PROJECTS

AgentProbe: ML Evaluation + Model Monitoring Infrastructure EVAL FRAMEWORK CI-INTEGRATED

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · MLflow-adjacent | github.com/venkatalakshmiparimala/AgentProbe

- 15-endpoint FastAPI backend, 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate), model routing across OpenAI/Anthropic/Groq/Ollama, CI-integrated regression harness with golden datasets, offline + online evaluation loops. Built to measure ML-powered workflows before and after deployment — the monitoring and regression testing layer for production models.

CrashClarity: Large-Scale ML Pipeline (104K Records) RANDOM FOREST FEATURE ENGINEERING

Python · scikit-learn · Random Forest · Logistic Regression · Feature Engineering · A/B Evaluation | Portfolio

- End-to-end ML pipeline over 104K+ UK road accident records — feature engineering across weather, road condition, time-of-day, and geographic variables; Random Forest + Logistic Regression ensemble (85%+ accuracy); A/B evaluation across model configurations; deployed interactive dashboard on GitHub Pages.

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia May 2026

GPA: 3.87 — coursework: ML pipelines, feature engineering, model evaluation, statistical modeling, data visualization, cloud platforms (AWS, GCP, Azure)

B.Tech Computer Science & Engineering JNTU Anantapur 2017

SKILLS

Languages: Python (8+ yrs), JavaScript, TypeScript, Java, SQL, R, Bash — FastAPI, Django, React.js, REST APIs

ML & Data: PyTorch, TensorFlow, XGBoost, scikit-learn, Random Forest, Logistic Regression, Isolation Forest, Feature Engineering, Feature Pipelines, Model Deployment, Model Serving, MLflow, A/B Testing, Anomaly Detection, Embeddings, HuggingFace, BERT, NLP, NumPy, Pandas, Tableau, Power BI

Backend & Data Infra: PostgreSQL, Redis, RabbitMQ (real-time event-driven), Apache Kafka, Celery, Temporal, Microservices, Multi-Tenant Architecture, Distributed Systems, Idempotency, ETL, Data Pipelines, OpenTelemetry, Prometheus, Grafana

Cloud & Infra: AWS (S3, EC2, RDS, Glue, Lambda, Step Functions, SageMaker), GCP (Vertex AI, Cloud Storage), Azure (Databricks, ADF, ADLS, AML), Docker, Kubernetes, Terraform, CI/CD (GitHub Actions), Serverless Architecture